A sharp order bound for an integral of the modified Bessel function

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Abstract

We prove Gaunt's conjectured lower bound for an integral of the modified Bessel function of the second kind, including the boundary at which the optimal order is zero. The bound is sharp among all fixed real orders. The key observation is strict convexity of $s \mapsto K_{\sqrt{s}}(x)$.

Task 1: Resolve MathDB problem 339406

Conjecture 2.4, equation (2.17), page 7 of [1], is the following theorem. We also prove its assertion about the best possible order.

Theorem 1. *Let $n < 1$ and $\nu \geq 1 - n + \sqrt{2(1-n)}$. Put*

$$m = \nu + n, \qquad d = \sqrt{m^2 - 2\nu - 1}, \qquad I(x) = \int_x^\infty t^{-\nu} K_m(t)\, dt.$$

Then, for every $x > 0$,

$$I(x) > x^{-\nu} K_d(x). \tag{1}$$

For any real β, put $\rho = |m - \beta|$. If $\rho < d$, then $K_\rho(x) < K_d(x)$ for every $x > 0$. If $\rho > d$, then $I(x) < x^{-\nu} K_\rho(x)$ for all sufficiently large x. Thus the optimal shifts are exactly $\beta = m - d$ and $\beta = m + d$, which give the same function and coincide when $d = 0$. For $n = 1$ the corresponding identity, valid for every real ν, is

$$\int_x^\infty t^{-\nu} K_{\nu+1}(t)\, dt = x^{-\nu} K_\nu(x). \tag{2}$$

Convexity in the squared order

We use the standard real-order representation

$$K_a(x) = \int_0^\infty e^{-x\cosh u} \cosh(au)\, du, \qquad a \in \mathbb{R}, \quad x > 0. \tag{3}$$

The recurrence and derivative identities used below are also recorded in Appendix B of [1].

Lemma 2. *For fixed $x > 0$, the function $F_x(s) = K_{\sqrt{s}}(x)$ is strictly increasing and strictly convex on $[0, \infty)$. Moreover,*

$$F_x'(1) = \frac{K_0(x)}{2x}. \tag{4}$$

Proof. For $u > 0$,

$$\cosh(u\sqrt{s}) = \sum_{j=0}^{\infty} \frac{u^{2j} s^j}{(2j)!}$$

is strictly increasing and strictly convex on $s \geq 0$; its first and second derivatives are positive, including at zero. Integration against the positive weight in (3) preserves both strict inequalities. Differentiation is justified locally uniformly in s: on compact order intervals, every required order derivative is dominated by a constant times a polynomial in u times $e^{-x\cosh u + Au}$ for some fixed A, an integrable function. Finally, integration by parts gives

$$2F_x'(1) = \int_0^\infty u \sinh u \, e^{-x\cosh u}\, du = \frac{1}{x}\int_0^\infty e^{-x\cosh u}\, du = \frac{K_0(x)}{x}.$$

The boundary term $ue^{-x\cosh u}$ vanishes at both endpoints. □

The derivative comparison

The assumptions imply

$$d^2 = (m-1)^2 - 2(1-n), \qquad 0 \leq d < m - 1 < \nu. \tag{5}$$

Indeed, $m - 1 \geq \sqrt{2(1-n)} > 0$ and $n < 1$. In particular,

$$m^2 - (d+1)^2 = 2(\nu - d) > 0.$$

First suppose $d > 0$. Apply strict convexity of F_x to the three ordered points $a = (d-1)^2 < b = (d+1)^2 < c = m^2$. The strictly increasing secant slopes give

$$\begin{aligned} K_m(x) &> K_{d+1}(x) + \frac{\nu - d}{2d}\big(K_{d+1}(x) - K_{d-1}(x)\big) \\ &= \frac{\nu + d}{2d} K_{d+1}(x) - \frac{\nu - d}{2d} K_{d-1}(x) \\ &= K_{d-1}(x) + \frac{\nu + d}{x} K_d(x), \end{aligned}$$

where $K_{-a} = K_a$ and $K_{d+1} - K_{d-1} = 2dK_d/x$ were used. If $d = 0$, then $m^2 = 1 + 2\nu > 1$. Strict convexity and (4) instead give

$$K_m(x) > F_x(1) + 2\nu F_x'(1) = K_1(x) + \frac{\nu}{x} K_0(x).$$

Consequently, in the full parameter range, including $d = 0$,

$$K_m(x) > K_{d-1}(x) + \frac{\nu + d}{x} K_d(x). \tag{6}$$

This proves the derivative inequality proposed in equation (2.18) of [1]; at $d = 0$ it gives its nonsingular form.

Proof of the integral bound and sharpness

Let $D(x) = I(x) - x^{-\nu}K_d(x)$. The identity $K_d' = -K_{d-1} - dK_d/x$ and (6) yield

$$D'(x) = x^{-\nu}\left(K_{d-1}(x) + \frac{\nu+d}{x}K_d(x) - K_m(x)\right) < 0.$$

Both terms defining D tend to zero at infinity, by exponential decay of the Bessel functions. Thus

$$D(x) = \int_x^\infty t^{-\nu}\left(K_m(t) - K_{d-1}(t) - \frac{\nu+d}{t}K_d(t)\right)dt > 0,$$

proving (1) with strictness.

For completeness we supply the optimality calculation suggested in the source. For each fixed real a, the standard expansion is

$$K_a(x) = \sqrt{\frac{\pi}{2}}e^{-x}x^{-1/2}\left(1 + \frac{4a^2-1}{8x} + O(x^{-2})\right).$$

Integration by parts for the first two terms, with the same order of remainder after integration, gives

$$\begin{aligned} I(x) &= \sqrt{\frac{\pi}{2}}e^{-x}x^{-\nu-1/2}\left(1 + \frac{(4m^2-1)/8 - \nu - 1/2}{x} + O(x^{-2})\right) \\ &= \sqrt{\frac{\pi}{2}}e^{-x}x^{-\nu-1/2}\left(1 + \frac{4d^2-1}{8x} + O(x^{-2})\right). \end{aligned}$$

It follows that

$$I(x) - x^{-\nu}K_\rho(x) = \sqrt{\frac{\pi}{2}}e^{-x}x^{-\nu-1/2}\left(\frac{d^2-\rho^2}{2x} + O(x^{-2})\right). \tag{7}$$

If $\rho > d$, the expression is eventually negative. If $0 \le \rho < d$, Lemma 2 gives $K_\rho < K_d$ pointwise. The only remaining orders are $\rho = d$, equivalent to the two stated shifts.

Finally, for $n = 1$, the identity $(x^{-\nu}K_\nu(x))' = -x^{-\nu}K_{\nu+1}(x)$ proves (2) after integration to infinity. It holds for all real ν, without imposing the $n < 1$ parameter condition.

Task 2: Classify the parameter range for sharp order bounds

Determine all real pairs (ν, n) for which a fixed real order r gives $\int_x^\infty t^{-\nu}K_{\nu+n}(t)\,dt \ge x^{-\nu}K_r(x)$ for every $x > 0$. Identify when the large-x obstruction $r^2 \le (\nu+n)^2 - 2\nu - 1$ is sufficient outside the range proved here.

Task 3: Quantify the positive remainder

Use the strictly positive convexity remainder in (6) to obtain explicit two-sided bounds for $D(x)$. Seek estimates that remain useful as $d \downarrow 0$ and $n \uparrow 1$, and that recover the first nonzero large-x term.

Task 4: Extend the sharp bound to exponential weights

Determine optimal Bessel orders and prefactors in bounds for $\int_x^\infty e^{\gamma t}t^{-\nu}K_{\nu+n}(t)\,dt$ when $\gamma < 1$. Establish which weighted analogues admit a convexity proof.

Task 5: Develop a general squared-order comparison principle

Identify other special-function families admitting a positive $\cosh(\sqrt{s}\,u)$ representation and useful order recurrences. Derive sharp integral comparisons from their secant and tangent inequalities, including parameter values where the secant degenerates.

References

[1] R. E. Gaunt, *Inequalities for some integrals involving modified Bessel functions*, Proc. Amer. Math. Soc. **147** (2019), 2937–2951. doi:10.1090/proc/14433. Author-linked manuscript: arXiv:1806.00524v2 (9 October 2018), Conjecture 2.4 and Remark 2.5, page 7; standard identities in Appendix B.